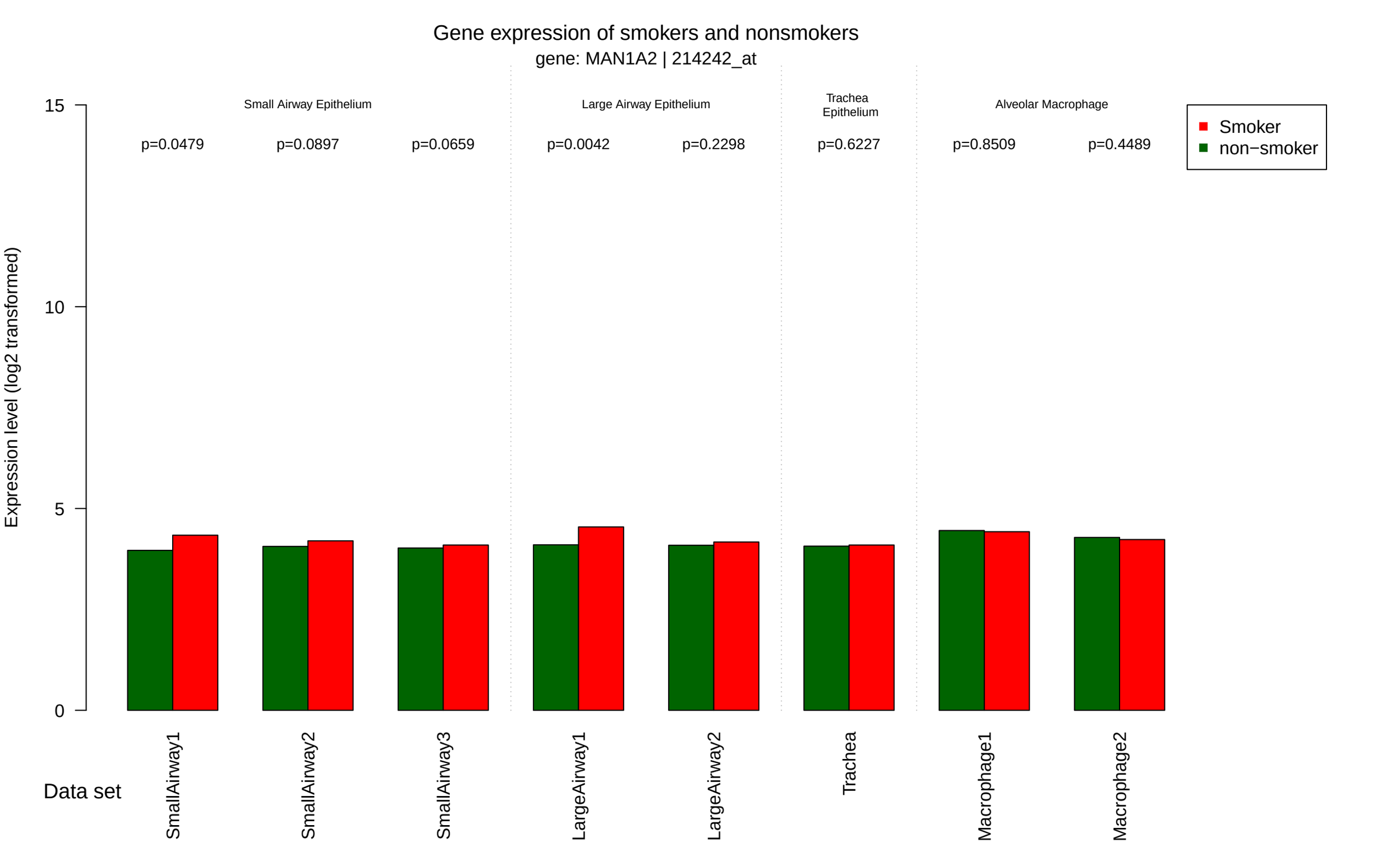
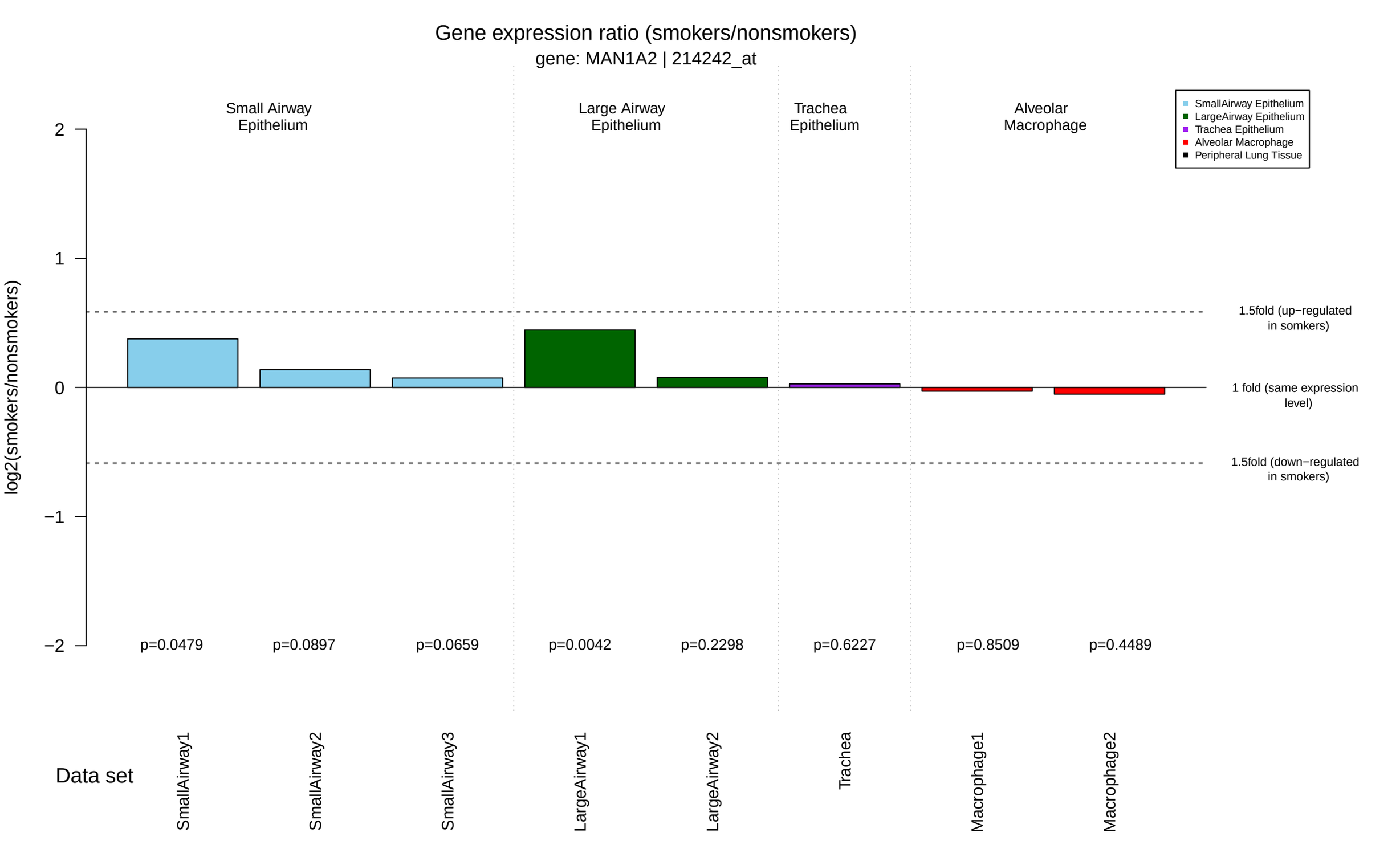
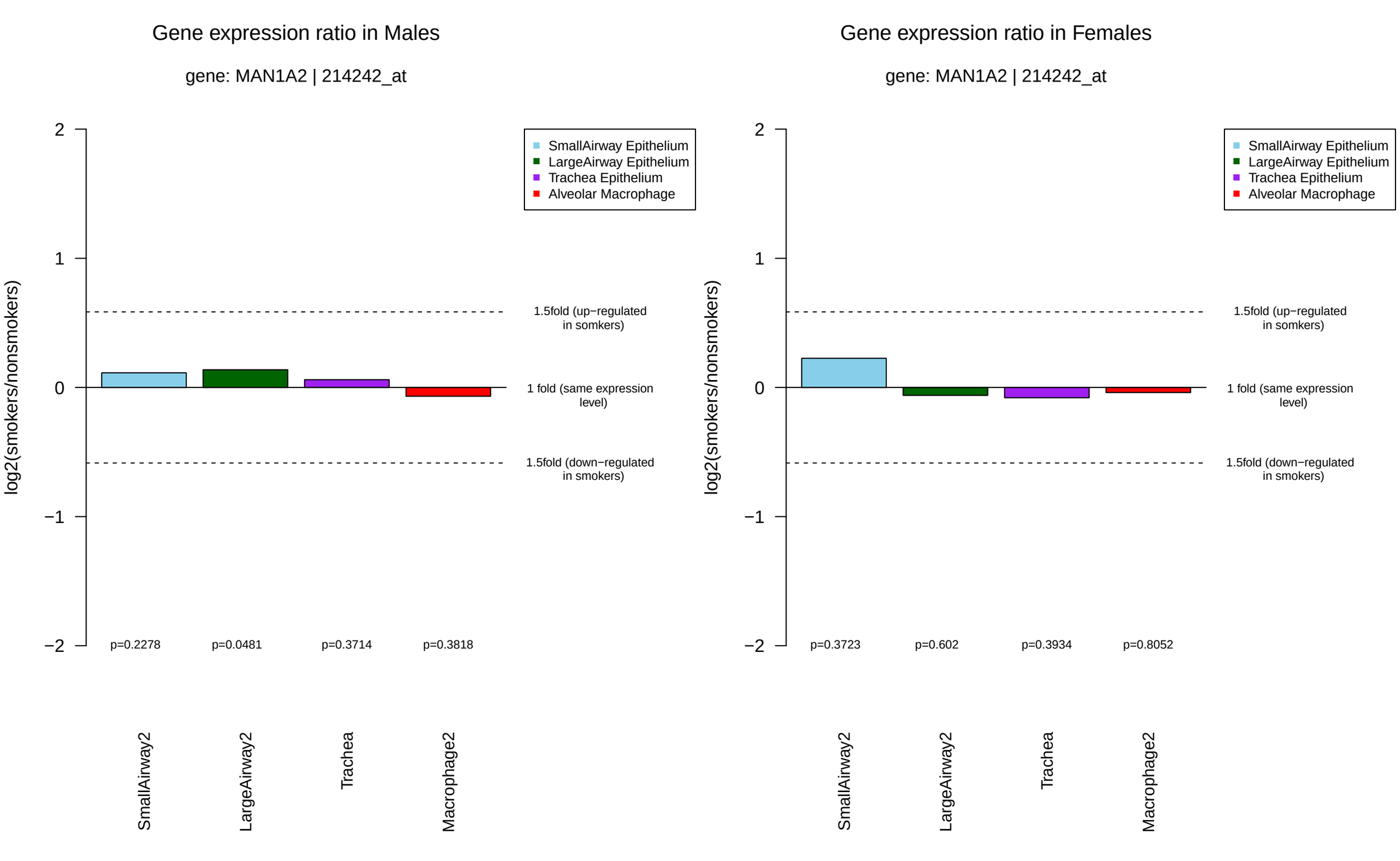
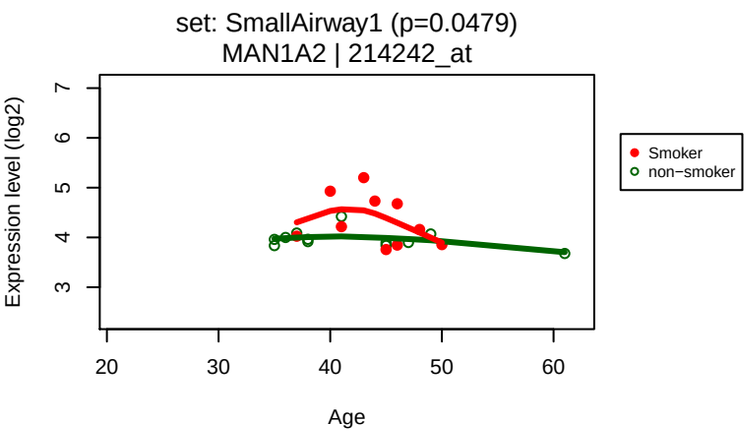
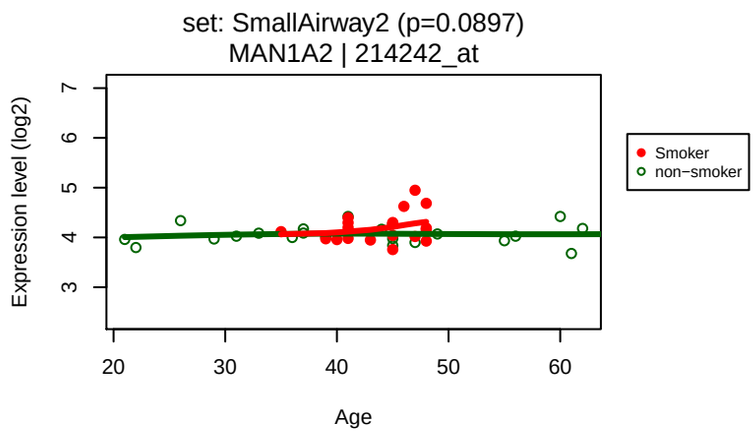
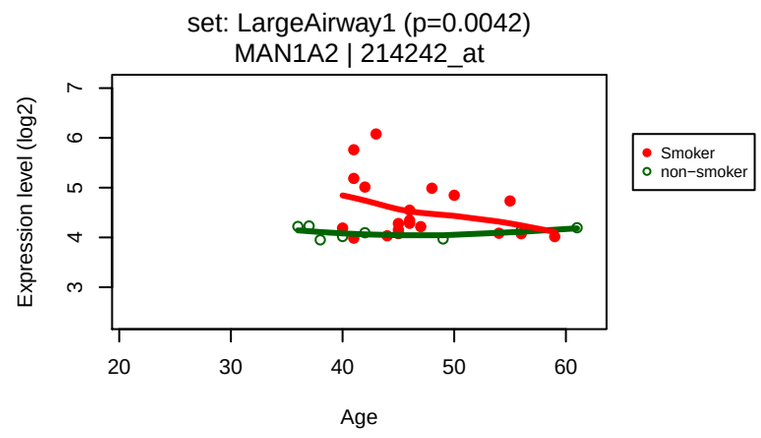

Gene expression of smokers and nonsmokers
gene: MAN1A2 | 214242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0479
p=0.0897
p=0.0659
p=0.0042
p=0.2298
p=0.6227
p=0.8509
p=0.4489
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

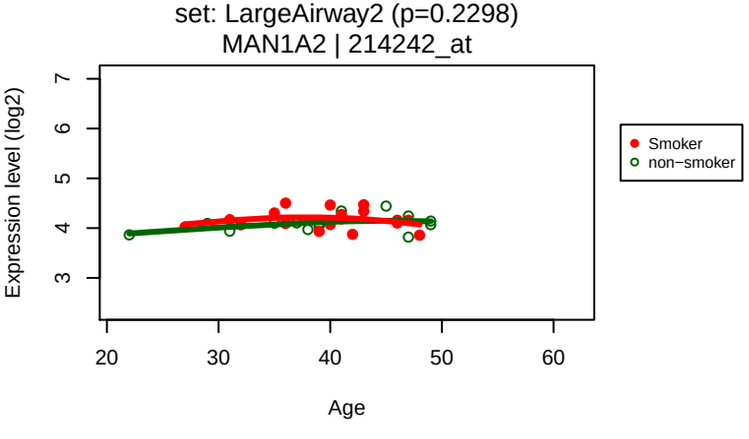

Gene expression ratio (smokers/nonsmokers)
gene: MAN1A2 | 214242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.0897
p=0.0659
p=0.0042
p=0.2298
p=0.6227
p=0.8509
p=0.4489
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

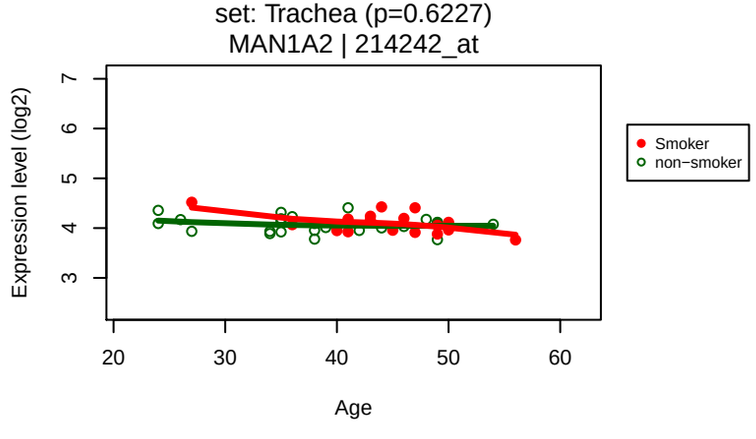

Gene expression ratio in Males
gene: MAN1A2 | 214242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2278
p=0.0481
p=0.3714
p=0.3818
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAN1A2 | 214242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3723
p=0.602
p=0.3934
p=0.8052
SmallAirway2
LargeAirway2
Trachea
Macrophage2

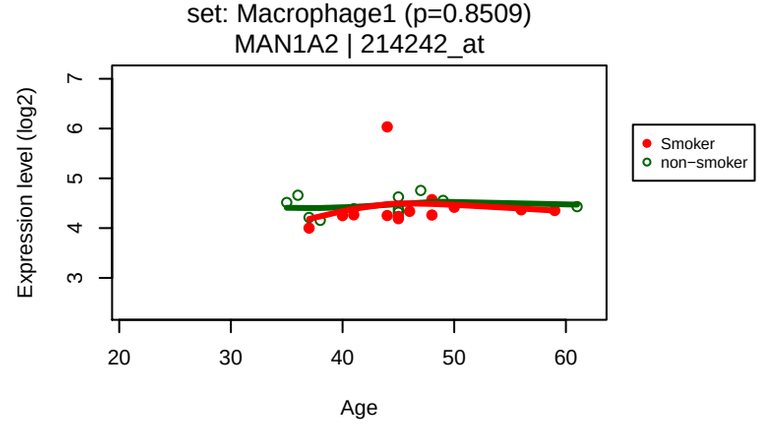

set: SmallAirway1 (p=0.0479)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker

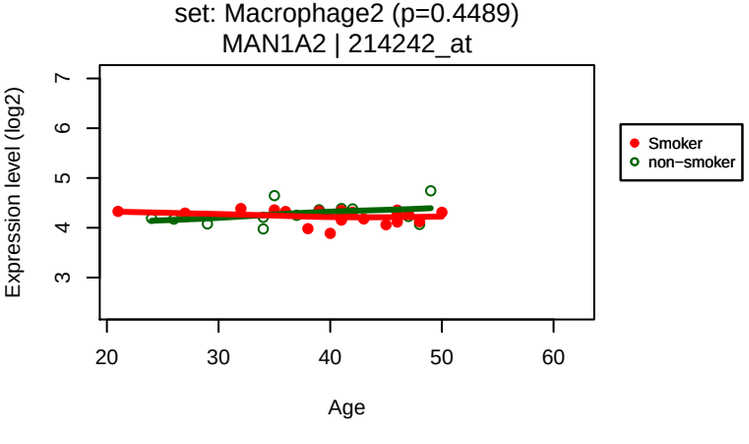

set: SmallAirway2 (p=0.0897)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker

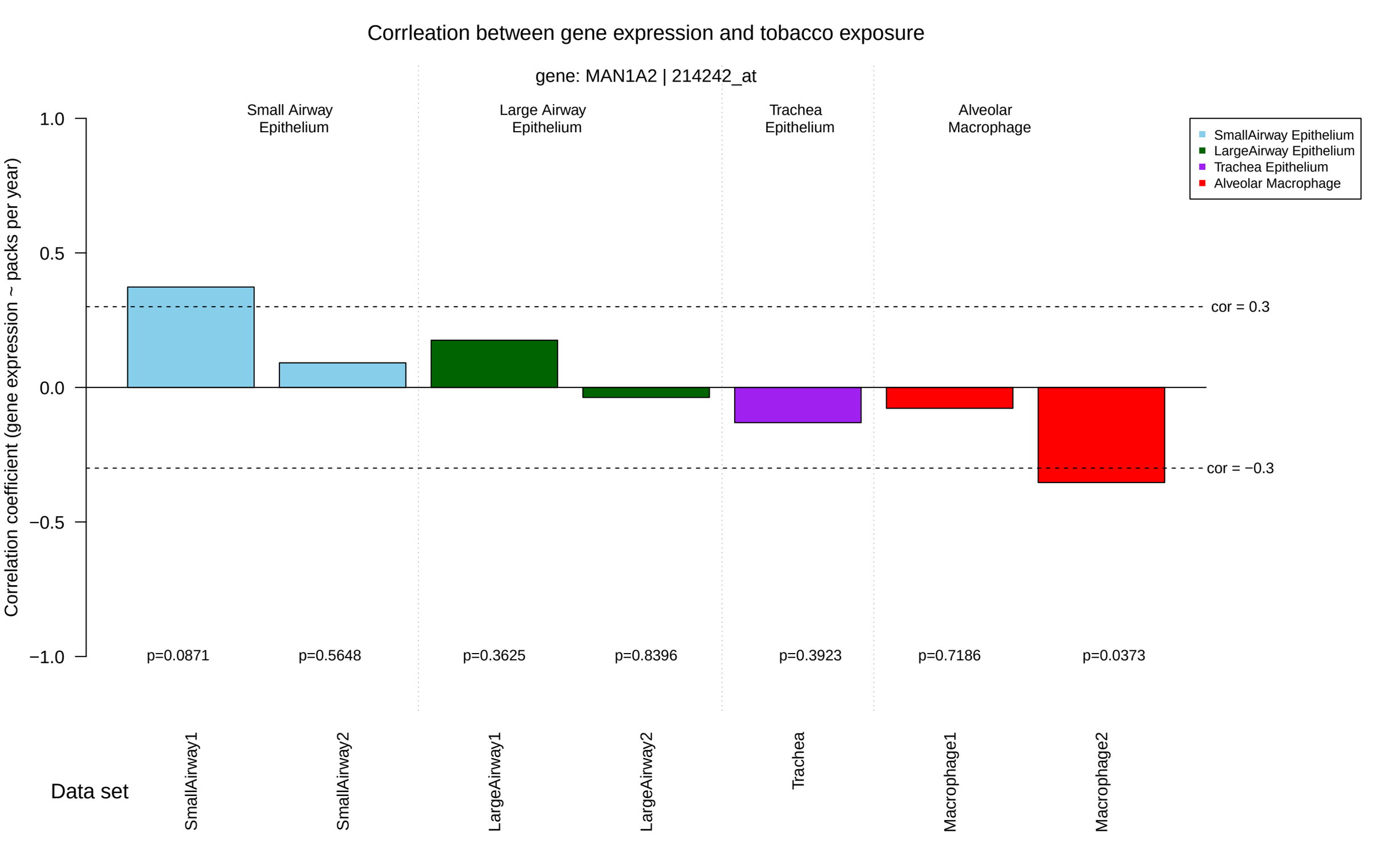

set: LargeAirway1 (p=0.0042)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2298)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6227)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8509)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4489)
MAN1A2 | 214242_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAN1A2 | 214242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0871
p=0.5648
p=0.3625
p=0.8396
p=0.3923
p=0.7186
p=0.0373
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2